

Answer all questions. (The right side numbers indicate marks.)

(Show the necessary steps whenever required)

1.
 - a) Find the Octal equivalent of hexadecimal number **AB.CD**. 2×6
 - b) Find the equivalent of Decimal **43** in Hexadecimal and BCD number system respectively
 - c) Find minimum number of 2-input NAND gates to implement the expression:

$$F = W(\overline{XY}) + Z(\overline{XY})$$
 - d) Convert **(0.28C)₁₆** to equivalent binary number.
 - e) Subtract two hexadecimal numbers **(86)₁₆ – (2A)₁₆**.
 - f) Perform **(19)₁₀ – (23)₁₀** using 2's complement technique.

2.
 - a) Convert the following SOP expression into simplified POS expression using K-Map only. 7

$$F = ABC\overline{B} + \overline{A}BC + \overline{A}\overline{B}D + \overline{B}\overline{C} + ABD + ABCD$$
 - b) ~~Convert the following SOP expression into simplified POS expression using K-Map only.~~ 5
~~$$F = ABC\overline{B} + \overline{A}BC + \overline{A}\overline{B}D + \overline{B}\overline{C} + ABD + ABCD$$~~
Find out the simplified SOP expression for $F(x, y, z, w) = \sum m(5, 6, 9, 11, 12, 15)$ using K-map only.

3.
 - a) Simplify the following Boolean expression using Boolean algebra (without using K-MAP). 6

$$(\overline{B}\overline{C} + D) + D + BC + AD$$
 - b) Express the following function as i) sum of minterms and ii) product of maxterms, using Boolean algebra (without using K-Map). 3+3

$$F(x, y, z) = (xy + z)(xz + y)$$

4.
 - a) Explain how a full adder can be implemented using half adders showing the necessary derivations and draw the block diagram for Full Adder using Half Adder. 6
 - b) Find out the Boolean expression for $A > B$ where A and B are two 4-bit binary numbers. Also draw the logic circuit for $A = B$. 3+3

5.
 - a) How many 4:1 multiplexers are required to design a 32:1 multiplexer (without using other logic gates)? 2
 - b) Design a 16:1 multiplexer using 8:1 multiplexers and/or necessary logic circuits. 4
 - c) Realize the Boolean expression $F(A, B, C, D) = \sum m(1, 2, 3, 5, 6, 8, 9, 13)$ using 8:1 MUX with A, B, C as selection lines. 6

2(b) Find out the simplified sop expression

for $F(x, y, z, w) = \pi M(5, 6, 9, 11, 12, 15)$ using

kmap only